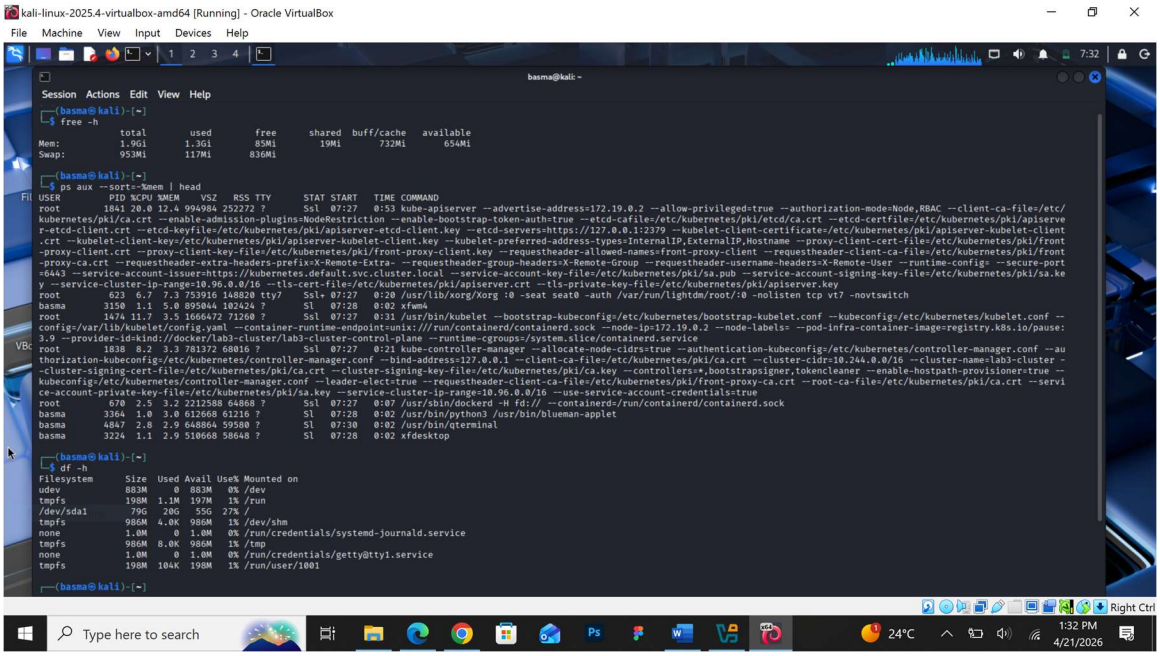
Name: Basma Mohamed Mostafa
Group: SE3

Lab 1

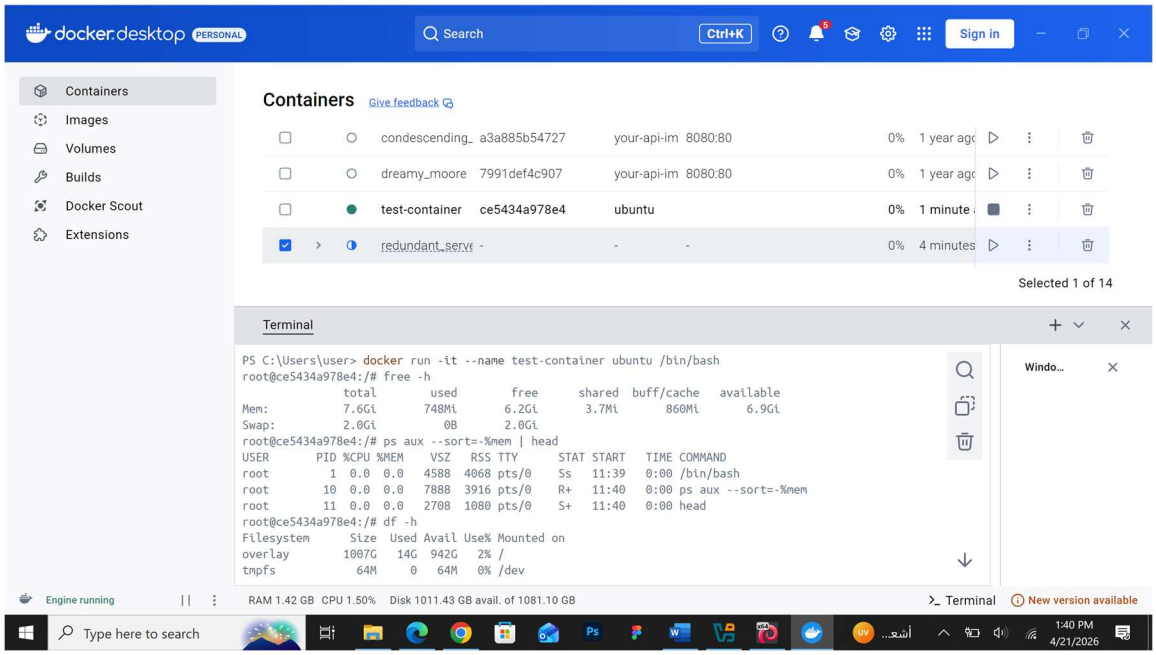
Deliverables:

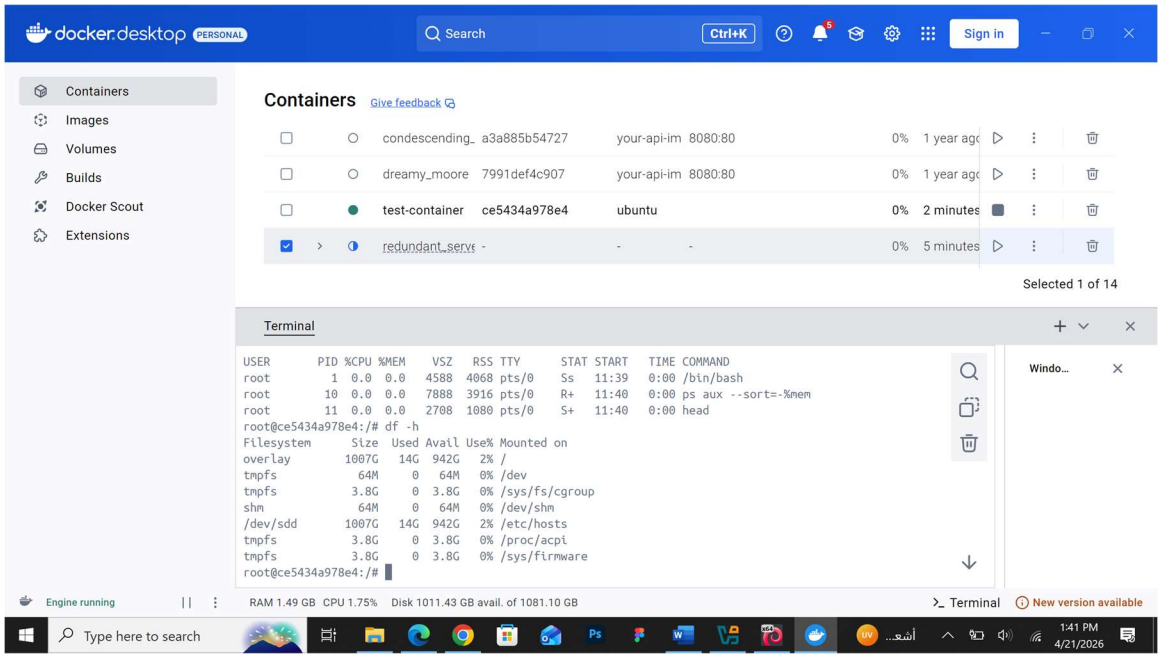
1. Screenshot comparisons of VM vs container resource usage.

   VM:



Container:


docker desktop PERSONAL
Search Ctrl+K
Sign in
Containers
Images
Volumes
Builds
Docker Scout
Extensions
Containers Give feedback
condescending_ a3a885b54727 your-api-im 8080:80 0% 1 year ago
dreamy_moore 7991def4c907 your-api-im 8080:80 0% 1 year ago
test-container ce5434a978e4 ubuntu 0% 1 minute
redundant_serve - - - 0% 4 minutes
Selected 1 of 14
Terminal
PS C:\Users\user> docker run -it --name test-container ubuntu /bin/bash
root@ce5434a978e4:/# free -h
total used free shared buff/cache available
Mem: 7.6Gi 748Mi 6.2Gi 3.7Mi 860Mi 6.9Gi
Swap: 2.0Gi 0B 2.0Gi
root@ce5434a978e4:/# ps aux --sort=-%mem | head
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.0 4588 4068 pts/0 Ss 11:39 0:00 /bin/bash
root 10 0.0 0.0 7888 3916 pts/0 R+ 11:40 0:00 ps aux --sort=-%mem
root 11 0.0 0.0 2708 1080 pts/0 S+ 11:40 0:00 head
root@ce5434a978e4:/# df -h
Filesystem Size Used Avail Use% Mounted on
overlay 1007G 14G 942G 2% /
tmpfs 64M 0 64M 0% /dev
Engine running
RAM 1.42 GB CPU 1.50% Disk 1011.43 GB avail. of 1081.10 GB
Terminal
New version available
Type here to search
1:40 PM
4/21/2026


docker desktop PERSONAL
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Containers
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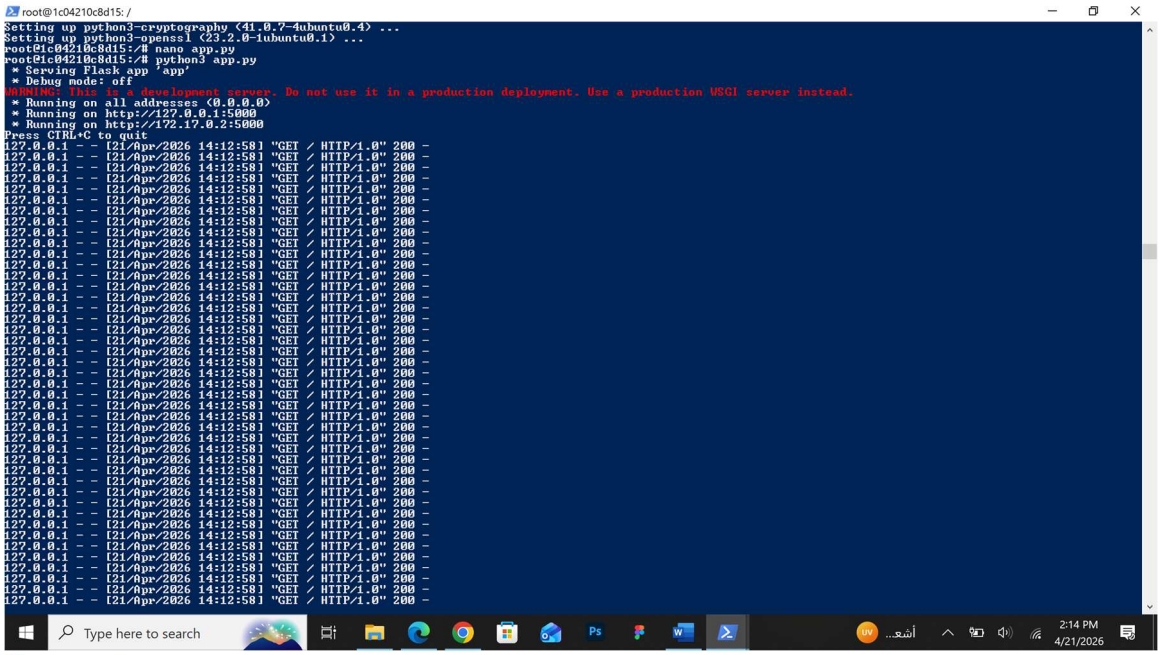
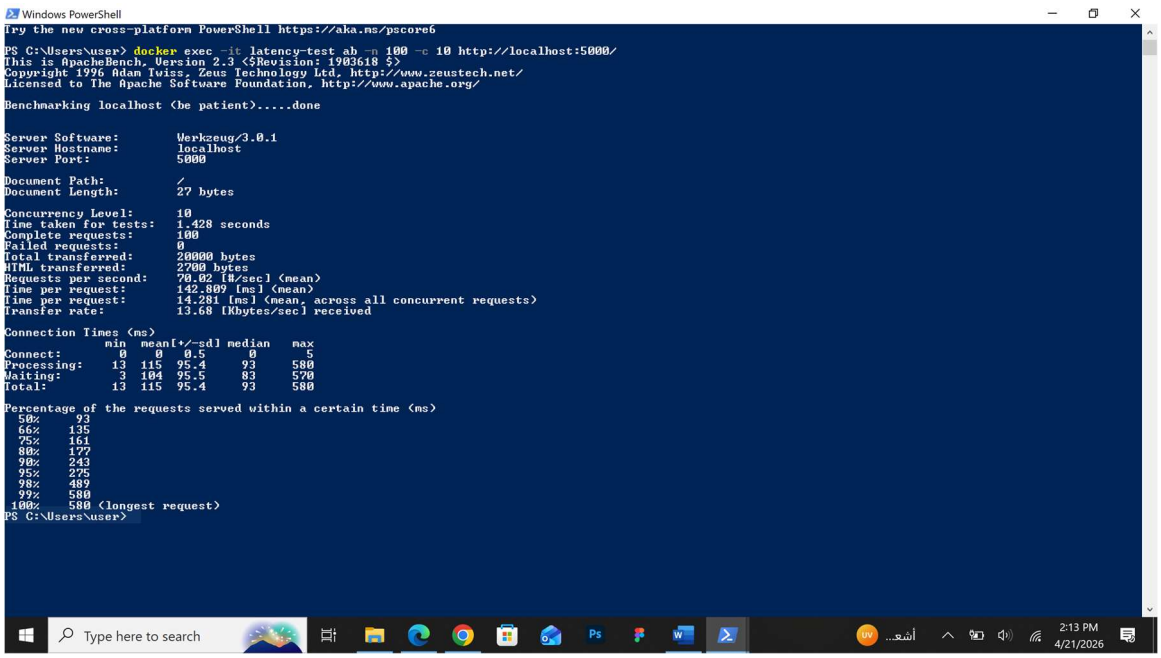
Containers Give feedback
condescending_ a3a885b54727 your-api-im 8080:80 0% 1 year ago
dreamy_moore 7991def4c907 your-api-im 8080:80 0% 1 year ago
test-container ce5434a978e4 ubuntu 0% 2 minutes
redundant_serve - - - 0% 5 minutes
Selected 1 of 14
Terminal
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.0 4588 4068 pts/0 Ss 11:39 0:00 /bin/bash
root 10 0.0 0.0 7888 3916 pts/0 R+ 11:40 0:00 ps aux --sort=-%mem
root 11 0.0 0.0 2708 1080 pts/0 S+ 11:40 0:00 head
root@ce5434a978e4:/# df -h
Filesystem Size Used Avail Use% Mounted on
overlay 1007G 14G 942G 2% /
tmpfs 64M 0 64M 0% /dev
tmpfs 3.8G 0 3.8G 0% /sys/fs/cgroup
shm 64M 0 64M 0% /dev/shm
/dev/sdd 1007G 14G 942G 2% /etc/hosts
tmpfs 3.8G 0 3.8G 0% /proc/acpi
tmpfs 3.8G 0 3.8G 0% /sys/firmware
root@ce5434a978e4:/#
Engine running
RAM 1.49 GB CPU 1.75% Disk 1011.43 GB avail. of 1081.10 GB
Terminal
New version available
Type here to search
1:41 PM
4/21/2026

2. Latency histogram of web service.





3. Short writeup: Which architecture (VM or container) is better for microservices? Why?

Containers are better for microservices. Their high density allows hundreds of services to run on a single host, and their fast startup times make them ideal for "Auto-scaling" and "Serverless" functions where instances need to be created and destroyed instantly based on traffic.